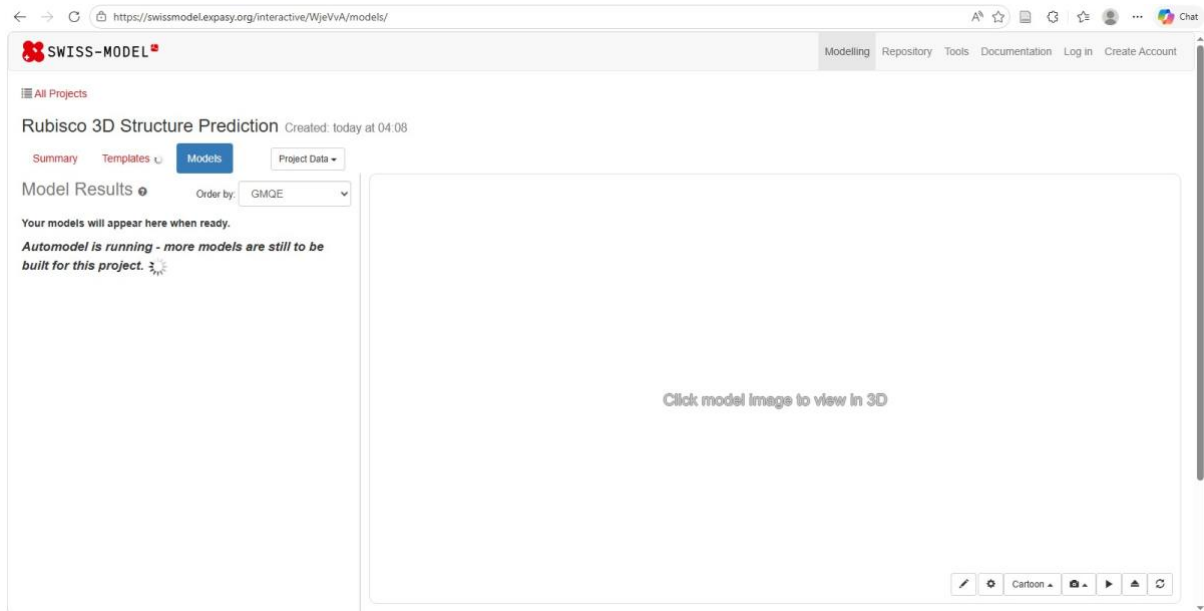


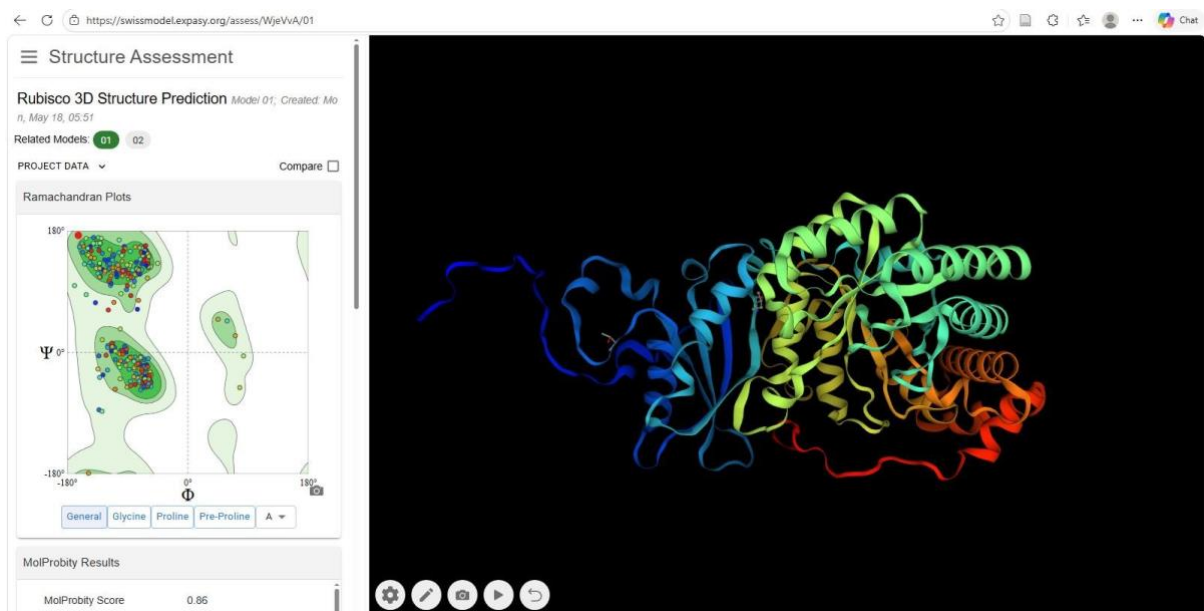
Task 3

Protein Structure Prediction

Step 1: Submitting Rubisco Sequence for 3D Modeling



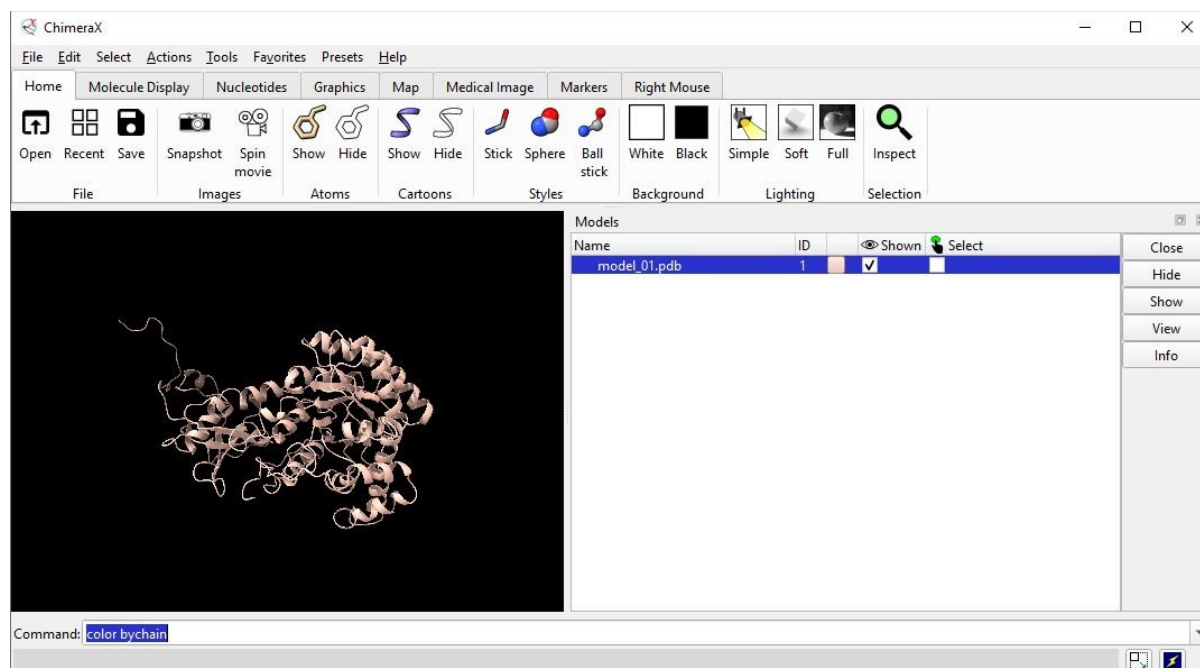
Step 2: Predicted 3D Structure and Visualization



3. Advanced Visualization using UCSF ChimeraX

To achieve high-resolution rendering and deeply analyze the tertiary structural elements, the predicted PDB model was imported into UCSF

ChimeraX. Full lighting presets were applied to capture accurate geometric depth and molecular shadows.



- **Structural Analysis:** The high-resolution rendering clearly showcases the intricate folding of the **Rubisco** monomer. The alphahelices (coils) and beta-sheets are tightly packed, forming the stable catalytic core required for carbon fixation during photosynthesis.
- **Command Line Validation:** The syntax execution line (color bychain) demonstrates the interactive capabilities used to distinguish molecular boundaries and individual polymer paths within the structural domain.

4. Conclusion

The 3D tertiary structure of the Rubisco enzyme was successfully predicted via SWISS-MODEL and advanced to a production-grade visual format using UCSF ChimeraX. The exceptional quality scores (GMQE and QMEAN) validate the computational reliability of this structural model. Gaining structural insights into this central photosynthetic enzyme provides a foundational framework for future agricultural biotechnology projects aimed at improving photosynthetic efficiency and crop yield traits.